Atlas 160: A Weighted K_4 Supporting Plane

Abstract

Let H be obtained from K_4 by adjoining a vertex on one clique edge and then attaching a pendant leaf to that new vertex. We prove, for every graphon of edge density $p \in [2/3, 1]$,

$$t(H, W) \geq p^2(2p-1)^2(3p-2).$$

This is the exact chromatic target and the full required interval. An elementary two-factor inequality reduces the graph to the signed expression $2t(J, W) - p\,t(K_4, W)$, where J is K_4 with a two-edge tail. We prove the needed strengthening of the earlier J bound directly. After conditioning at the tail-bearing clique vertex, upper and lower triangle estimates in its link give a piecewise scalar function of d and $A = T_W d$. A sharp supporting plane integrates exactly because $\int A = \int d^2$. All remaining signs are certified by explicit rational Bernstein expansions.

1 The graph, target, and graph reduction

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and measurable. Write

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y),$$

and

$$A(x) = (T_W d)(x) = \int_{\Omega} W(x,y) d(y)\,\mathrm{d}\mu(y).$$

Take clique vertices $0, 1, 2, 3$, a page vertex z, and a leaf u. The graph H has all six edges of the K_4, the edges $1z, 2z$, and zu. Its chromatic polynomial and target are

$$\chi_H(x) = x(x-1)^2(x-2)^2(x-3), \tag{1}$$

$$\Phi_H(p) = p^2(2p-1)^2(3p-2). \tag{2}$$

Indeed, after colouring the K_4, the page has $x-2$ choices and the leaf has $x-1$ choices. Thus $\chi(H) = 4$ and the required interval is $p \in [2/3, 1]$.

For $x, y \in \Omega$, put

$$B(x,y) = \int_{\Omega} W(x,z)W(y,z)d(z)\,\mathrm{d}\mu(z).$$

Let $\kappa_4(x_0, x_1, x_2, x_3)$ be the product of the six clique-edge factors. Integrating first over u and z gives

$$t(H, W) = \int_{\Omega^4} \kappa_4 B(x_1, x_2)\,\mathrm{d}\mu^4.$$

The identity

$$rs = r + s - 1 + (1-r)(1-s)$$

and nonnegativity imply

$$B(x,y) \geq A(x) + A(y) - p.$$

Consequently, by symmetry among the four clique coordinates,

$$t(H,W) \geq 2J - pt_4, \tag{3}$$

where $t_4 = t(K_4, W)$ and

$$J = \int_{\Omega^4} \kappa_4 A(x_1)\, \mathrm{d}\mu^4.$$

The latter is the density of K_4 with an otherwise disjoint path of length two attached at one clique vertex. The bound for J alone from the earlier K_4 two-edge-tail theorem is not enough in (3); the signed combination must be kept intact.

2 The rooted scalar reduction

Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\, \mathrm{d}\mu(y)\, \mathrm{d}\mu(z).$$

When $d(x) > 0$, let

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)}\, \mathrm{d}\mu(y)$$

and regard W as a graphon W_x on this probability space. Its edge density is

$$z_x = \frac{\tau(x)}{d(x)^2}.$$

Rooting the K_4 at the vertex carrying the A weight gives the exact identity

$$2J - pt_4 = \int_{\Omega} d(x)^3 (2A(x) - p) t(K_3, W_x)\, \mathrm{d}\mu(x). \tag{4}$$

We use three elementary facts. First, triangle Goodman says that a graphon of edge density z has triangle density at least

$$f(z) = z(2z-1)_+.$$

This function is nondecreasing. Second,

$$\tau(x) \geq (2A(x) - p)_+. \tag{5}$$

Indeed, apply $rs \geq r + s - 1$ to $W(x,y), W(x,z)$ and multiply by $W(y,z)$. Third, every triangle density is at most 1.

For $d > 0$, put $s = 2a - p$ and define

$$\mathcal{F}_p(d,a) = \begin{cases} d^3 s, & s < 0, \\ 0, & 0 \leq s \leq d^2/2, \\ \dfrac{s^2(2s - d^2)}{d}, & s \geq d^2/2. \end{cases} \tag{6}$$

At $d = 0$, set $\mathcal{F}_p(0,0) = 0$. If $s < 0$, the upper bound $t(K_3, W_x) \le 1$ reverses in the correct direction after multiplication by s. If $0 \le s \le d^2/2$, nonnegativity suffices. In the last region, (5), monotonicity of f, and Goodman give

$$d^3 s\, t(K_3, W_x) \ge d^3 s f(s/d^2) = \frac{s^2(2s - d^2)}{d}.$$

The actual rooted variables satisfy $s \le d^2$ whenever $s \ge 0$, since $\tau \le d^2$; hence the argument of f lies in $[0, 1]$. Thus (4) implies

$$2J - pt_4 \ge \int_\Omega \mathcal{F}_p(d(x), A(x))\, \mathrm{d}\mu(x). \tag{7}$$

The coarse feasible information that will make the supporting plane integrate is

$$0 \le A(x) \le d(x), \qquad A(x) \ge d(x) + p - 1, \tag{8}$$

and

$$\int d = p, \qquad \int A = \int d^2. \tag{9}$$

For the lower pointwise bound in (8), apply $rs \ge r + s - 1$ to $W(x, y), d(y)$ and integrate. The moment identity follows from symmetry and Tonelli.

3 The supporting plane

Put

$$T_p = p^2(2p - 1)^2(3p - 2), \tag{10}$$
$$\gamma_p = 4p(2p - 1)(5p - 3), \tag{11}$$
$$\beta_p = p(2p - 1)(30p^2 - 15p - 2), \tag{12}$$

and define

$$L_p(d, a) = T_p + \beta_p(d - p) + \gamma_p(a - d^2). \tag{13}$$

These are exactly the value and the derivatives, in coordinates $(d, a - d^2)$, of the active expression in (6) at $(d, a) = (p, p^2)$.

Lemma 3.1 (Sharp scalar support)**.** *Let $p \in [2/3, 1]$. If $0 \le d \le 1$ and*

$$\max\{0, d + p - 1\} \le a \le d,$$

then, on every portion of the region relevant to (6)*,*

$$\mathcal{F}_p(d, a) \ge L_p(d, a). \tag{14}$$

Proof. If $d = 0$, feasibility gives $a = 0$ and

$$L_p(0, 0) = -4p^2(2p - 1)(6p^2 - 2p - 1) \le 0.$$

Assume $d > 0$. Notice that $\gamma_p > 0$.

We first handle $a < p/2$. Here the scalar lower bound is $d^3(2a-p)$. Its difference from $L_p(d,a)$ is affine in a on the interval

$$\max\{0, d+p-1\} \le a \le \min\{d, p/2\}.$$

It is therefore enough to check the four possible boundary faces

$$a = 0, \qquad a = d+p-1, \qquad a = d, \qquad a = p/2.$$

Write $p = (u+2)/3$, with $0 \le u \le 1$, and parameterize their respective d intervals by

$$d = (1-p)v, quadd = 1-p+\frac{p}{2}v, quadd = \frac{p}{2}v, quadd = \frac{p}{2}+(1-p)v, qquad 0 \le v \le 1.$$

In the native tensor-product Bernstein bases, the four residuals have the following integer coefficient matrices after multiplication by the positive denominators displayed on the left:

$$7290 \left| \begin{pmatrix} 1440 & 1200 & 1040 & 780 \\ 4032 & 3420 & 2976 & 2610 \\ 9828 & 8595 & 7614 & 6858 \\ 21708 & 19764 & 18036 & 16524 \\ 44712 & 42606 & 40500 & 38394 \\ 87480 & 87480 & 87480 & 87480 \end{pmatrix} \right.$$

$$116640 \left| \begin{pmatrix} 12480 & 7200 & 2560 & 0 & 3840 \\ 41760 & 29304 & 18144 & 9648 & 9216 \\ 109728 & 80406 & 54396 & 32724 & 22032 \\ 264384 & 197640 & 139050 & 88857 & 55080 \\ 614304 & 468504 & 341172 & 231093 & 147744 \\ 1399680 & 1093500 & 826200 & 594135 & 408240 \end{pmatrix} \right.$$

$$116640 \left| \begin{pmatrix} 23040 & 17280 & 12160 & 6960 & 3840 \\ 64512 & 48816 & 34848 & 21600 & 12096 \\ 157248 & 119880 & 86688 & 56268 & 32832 \\ 347328 & 266814 & 195588 & 131706 & 81000 \\ 715392 & 554040 & 412128 & 286983 & 186624 \\ 1399680 & 1093500 & 826200 & 594135 & 408240 \end{pmatrix} \right.$$

$$4860 \left| \begin{pmatrix} 160 & 80 & 160 \\ 504 & 276 & 384 \\ 1368 & 891 & 918 \\ 3375 & 2619 & 2295 \\ 7776 & 6966 & 6156 \\ 17010 & 17010 & 17010 \end{pmatrix} \right. .$$

Every coefficient is nonnegative. Since the Bernstein basis functions are nonnegative and sum to one, all four boundary residuals are nonnegative. Affineness in a proves the whole negative-coefficient region.

Now put

$$a_0 = \frac{p}{2} + \frac{d^2}{4}.$$

At this switching point, direct expansion gives

$$-L_p(d, a_0) = p(2p-1)S_p(d), \tag{15}$$

where

$$S_p(d) = (15p - 9)d^2 + (-30p^2 + 15p + 2)d + 24p^3 - 18p^2 + 2p.$$

Its leading coefficient is positive, while

$$\operatorname{disc}_d S_p = -(3p - 2)^2(60p^2 - 36p - 1) \le 0.$$

The last factor is positive at $p = 2/3$ and increasing thereafter. Hence $S_p(d) \ge 0$. Since L_p is increasing in a, this proves $0 \ge L_p(d, a)$ throughout $p/2 \le a \le a_0$, which is the middle region of (6).

It remains to treat $a \ge a_0$. Let

$$G_{p,d}(a) = d\big(\mathcal{F}_p(d, a) - L_p(d, a)\big),$$

using the active expression from (6). This is a cubic in a with positive leading coefficient. Exact calculation gives

$$\operatorname{disc}_a G_{p,d} = -256d^2p(d - p)^2(2p - 1)K(p, d), \tag{16}$$

where

$$\begin{aligned} K(p, d) = \; & -20d^5p + 12d^5 - 10d^4p^2 + 9d^4p - 2d^4 - 24d^3p^3 + 24d^3p^2 - 6d^3p \\ & + 17912d^2p^4 - 30476d^2p^3 + 17222d^2p^2 - 3231d^2p \\ & - 23552dp^5 + 34064dp^4 - 12080dp^3 - 1908dp^2 + 1188dp \\ & + 31104p^6 - 62208p^5 + 46008p^4 - 15228p^3 + 2160p^2 - 108p. \end{aligned}$$

To certify its sign, put $p = (u + 2)/3$ and $d = v$. In the degree-$(6, 5)$ tensor-product Bernstein basis,

$$K((u + 2)/3, v) = \frac{1}{72900} \sum_{i=0}^{6} \sum_{j=0}^{5} M_{ij} B_i^6(u) B_j^5(v),$$

where

$$M = \begin{pmatrix} 194400 & 76800 & 20940 & 23580 & 75000 & 61800 \\ 729000 & 326480 & 117745 & 97125 & 248960 & 369400 \\ 2468880 & 1239888 & 554634 & 403722 & 763176 & 1335240 \\ 7479540 & 4182732 & 2230929 & 1609389 & 2283120 & 3856680 \\ 20586960 & 12747888 & 7847280 & 5863104 & 6746328 & 9999720 \\ 52488000 & 35630280 & 24545025 & 19200645 & 19530720 & 24931800 \\ 125971200 & 92612160 & 69655950 & 57058830 & 54733320 & 61965000 \end{pmatrix}.$$

Every entry is positive, so $K(p, d) > 0$ on the entire unit box.

If $d \ne p$, equation (16) says that the cubic has exactly one real root. Equation (15) is strict in this case: for $p > 2/3$ the quadratic discriminant is negative, while at $p = 2/3$,

$$S_{2/3}(d) = \frac{(3d - 2)^2}{9},$$

whose only zero is $d = p$. Thus $G_{p,d}(a_0) > 0$. A real cubic with positive leading coefficient and exactly one real root is positive to the right of that root, so $G_{p,d}(a) > 0$ for every $a \ge a_0$.

If $d = p$, direct factorization gives instead

$$G_{p,p}(a) = 4(a - p^2)^2(4a + 7p^2 - 6p).$$

At $a \ge a_0$, the last factor is at least $4p(2p - 1) > 0$. Hence this expression is nonnegative too. Division by $d > 0$ completes the active case and the proof. □

4 The graphon theorem and Atlas identification

Theorem 4.1 (Atlas 160)**.** *Every graphon W of edge density $p \in [2/3, 1]$ satisfies*

$$\boxed{t(H, W) \geq p^2(2p-1)^2(3p-2) = \Phi_H(p).}$$

Thus NetworkX Atlas graph 160*, with graph6 code* `E~@g`*, is positive on its complete required interval.*

Proof. Combine (3), (7), and Lemma 3.1, then use (9):

$$\begin{aligned} t(H, W) &\geq \int \mathcal{F}_p(d, A) \\ &\geq T_p + \beta_p \left(\int d - p \right) + \gamma_p \left(\int A - \int d^2 \right) \\ &= T_p. \end{aligned}$$

Equation (2) identifies this with the chromatic target. □

At $p = 2/3$ the target vanishes, and at $p = 1$ the graphon is 1 almost everywhere. The proof nevertheless includes both endpoints. Every integrand is a bounded measurable finite product, so Tonelli–Fubini applies on an arbitrary probability space. Zero-degree fibres are handled before division. No regularity, minimum-degree, finite-step, or injectivity hypothesis is used.

The graph identification, chromatic target, graph reduction, supporting plane, discriminants, factorizations, and all 144 displayed Bernstein coefficients were reconstructed in exact rational arithmetic.

A What the Lean formalization actually proves

Theorem 4.1 is formalized in full, on $p \in [2/3, 1]$, so the closing sentence above is superseded. The development is `lean/Taeyoung/Methods/Atlas160/`, the catalogue theorem is `satisfiesLowerBound_160`, and every declaration depends on exactly `[propext, Classical.choice, Quot.sound]`. Atlas 160 is `verified`.

Sections 1 and 2 are followed as written. Lemma 3.1 is not: of the 144 Bernstein coefficients it rests on, the formalization uses 42, and those only on one face.

A.1 The active region is one factorization

The discriminant (16), the 7×6 integer matrix certifying $K(p, d) > 0$, and the argument that a real cubic with positive leading coefficient and a single real root is positive to the right of it, are all replaced. In the coordinate $w = 4a - 2p - d^2$ the cubic is factored explicitly: with $M = 11p - 6$ and

$$R = 2d(3p^2 - 7p + 3) + p(27p^2 - 26p + 6),$$

$$M^3 \cdot 4\big(s^2 w - d\, L_p\big) = (Mw - R)^2\big(Mw + 2R + 2Md^2\big) + M(d-p)^2\,\mathrm{LIN} \cdot w + (d-p)^2\,\mathrm{CON},$$

a polynomial identity, where R/M is the tangent line at $d = p$ of the double root of (15). The two remainder coefficients are nonnegative for elementary reasons: $\mathrm{LIN} = \big(Md + 23p^2 - 34p + 12\big)^2 + 2(829p^4 - 1324p^3 + 646p^2 - 60p - 18)$ is a square plus a quartic in p alone, and $\mathrm{CON} = Ad^2 + Bd + C$ has $A \leq 0$, so $d^2 \leq d$ and then $d \geq p/2$ reduce it to $(A + B)p/2 + C$, again a polynomial in p alone. Every one-variable fact left over has all coefficients nonnegative after $p = 2/3 + u$.

This is the same device that carries the K_4 two-edge-tail row in `Methods/K4Tail/Scalar.lean`; only the coefficients differ.

A.2 The negative region needs two faces, not four

The residual $d^3 s - L_p$ is affine in a with slope $2d^3 - \gamma_p$, and the feasible interval (8) supplies both $a \le p/2$ and $a \ge d + p - 1$. So the face $a = p/2$ settles the slope-nonpositive case and the face $a = d + p - 1$ the slope-nonnegative one; the faces $a = 0$ and $a = d$ are never needed.

The first face is $p(2p-1)Q_1(d)$ with Q_1 a quadratic of positive leading coefficient and nonpositive discriminant, so no certificate is required. Only the second needs one, and the slope hypothesis confines it: $2d^3 \ge \gamma_p$ forces $d \ge 1/2$ and $p \le 5/6$, and on that box a degree-$(6, 5)$ Bernstein expansion has all 42 coefficients positive. On the full box the same expansion does not, because $d^3 s - L_p$ vanishes at $(p, d) = (1, 1)$ — a corner the slope hypothesis excludes.

A.3 Two pointwise facts the note leaves implicit

The reduction (3) is carried out with only two coordinates moving. Peeling the leaf and the page from the six-fold integral leaves $\iint \mathrm{pairK}(a_0, a_1)\, B(a_0, a_1)$, where pairK is the K_4 weight with the two page-bearing coordinates fixed; it is nonnegative and symmetric, so the monotonicity step is a two-level `integral_mono` and the identification of the two tail placements is one Fubini swap rather than a relabelling of a four-fold integral.

Two estimates on κ_4 are used that the note states only in passing. The $s < 0$ branch of (6) needs $\kappa_4(x) \le d(x)^3$, which is $t(K_3, W_x) \le 1$ off the degenerate set and the almost-everywhere vanishing of $W(x, \cdot)$ on it; and the lower feasibility bound of (8) is $(1 - W(x, y))(1 - d(y)) \ge 0$ integrated. Both are proved, so the degenerate fibres need no convention.